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BLOCKCHAIN EXPERIMENT NO: 4

Aim - Creation of account in MetaMask, and transfer of amount/cryptocurrency from one account to another.

Theory:

1. What is MetaMask?

MetaMask is a non-custodial cryptocurrency wallet, available as a browser extension and a mobile app, that lets a user generate and manage Ethereum-compatible accounts, hold ETH and other tokens, and interact with decentralized applications and blockchains directly from the browser. Being non-custodial means the wallet's private keys are generated and stored locally on the user's device, encrypted with a password and recoverable through a 12-word secret recovery phrase, rather than being held by any company or exchange. The account holder therefore has full control of, and full responsibility for, their funds.

2. Account Creation in MetaMask

When a new MetaMask account is created, the wallet generates a public/private key pair. The private key is used to sign transactions and must never be shared, while the public key is hashed to derive the account's public wallet address, for example 0x37F8f...41234, which is the identifier others use to send funds to that account. A single MetaMask installation can hold multiple accounts, all derived from the same seed phrase or imported separately, letting one user manage several addresses from a single interface.

3. Testnets and Test Tokens

Since real ETH has monetary value, blockchain transactions during development and learning are usually performed on a testnet, which is a separate blockchain that mimics the mainnet's behaviour but uses worthless test tokens. In this experiment the Sepolia Testnet was used, and the account was funded with test SepoliaETH obtained from a Sepolia faucet, allowing transactions to be sent and confirmed without any real financial cost or risk.

4. How a Transaction Works

A transaction in MetaMask consists of a sender address, a recipient address, an amount to transfer, and a gas fee paid to the network for processing it. When a user initiates a Send transaction, the following steps occur:

1. The sender specifies the recipient's address and the amount to transfer.
2. MetaMask estimates the network fee (gas) required to process the transaction, based on current network congestion.
3. The user reviews the transaction details, namely sender, recipient, network, amount and fee, and confirms it by signing with their private key.
4. The signed transaction is broadcast to the network, picked up by validators, and included in a block.
5. Once mined, the recipient's balance is updated and the sender's balance is reduced by the amount sent plus the gas fee.

This process demonstrates the core blockchain principle of peer-to-peer value transfer without an intermediary such as a bank: the transaction is validated and recorded directly on the distributed ledger, and is publicly verifiable by anyone through a block explorer.

Software Requirements:

- MetaMask browser extension
- Web browser (Chrome / Brave / Firefox)
- Google Cloud Web3 Sepolia faucet for obtaining test tokens
- Sepolia Etherscan block explorer for verification
- An active internet connection

Procedure:

1. Install the MetaMask extension and create a new wallet account, securely storing the 12-word secret recovery phrase.
2. Switch the network from Ethereum Mainnet to the Sepolia Testnet.
3. Copy the account's public wallet address.
4. Open the Google Cloud Web3 Sepolia faucet, paste the wallet address and request test SepoliaETH.
5. Verify that the funded balance appears in the MetaMask wallet.
6. Select the Send option, enter the recipient address and specify the amount to transfer.
7. Review the confirmation screen showing sender, recipient, network, amount and estimated network fee, then confirm the transaction.
8. Observe the pending status while the transaction is broadcast and mined.
9. Verify the updated account balance and the completed entry in the Activity tab.
10. Open the transaction on Sepolia Etherscan to verify it on the public ledger.

Transaction Details:

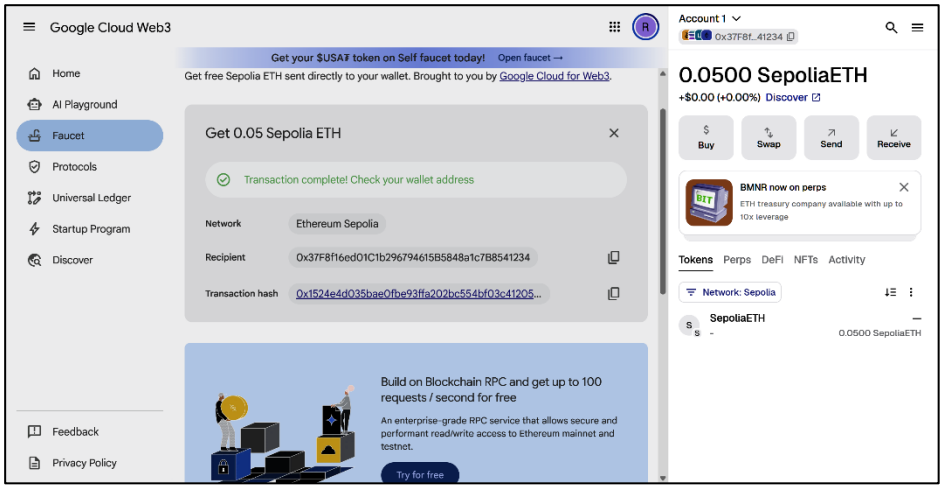
Parameter	Value
Network	Sepolia Testnet
Sender account	Account 1
Sender address	0x37F8f16ed01C1b296794615B5848a1c7B8541234
Recipient address	0x78d2Cd1Ce519284f40c174D40c86e2546c9D50FA
Amount transferred	0.01 SepoliaETH
Network fee (gas)	0.0001 SepoliaETH
Balance before transfer	0.0500 SepoliaETH
Balance after transfer	0.0399 SepoliaETH
Transaction hash	0x4028922aa800b11067c30e61aa74feb7eb795f908de0e8569427211657ceffa3
Block number	11658562
Status	Success / Confirmed
Timestamp	Sep-08-2026, 03:33:24 AM UTC

Output:

1. Account Funded and Transfer Initiated

A MetaMask account was created on the Sepolia Testnet and funded with 0.0500 SepoliaETH from the Google Cloud Web3 Sepolia faucet. The Send option was then selected, the recipient address 0x78d2C...D50FA was entered and an amount of 0.01 SepoliaETH was specified for transfer.

Output:



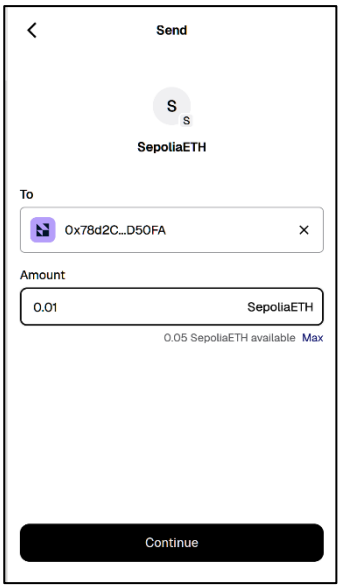
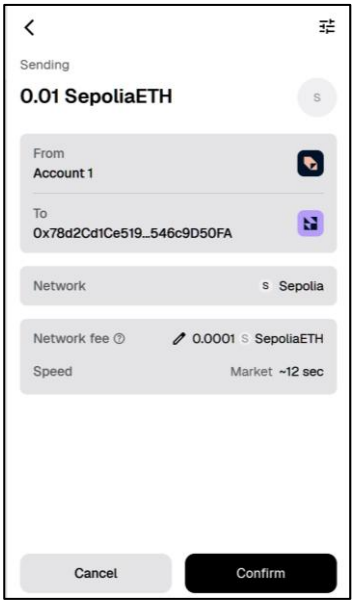


Fig 1 (left): Faucet transaction complete, wallet showing 0.0500 SepoliaETH. Fig 2 (right): Send screen with recipient address and amount entered.

2. Confirming and Broadcasting the Transaction

MetaMask displayed a confirmation screen summarizing the sender (Account 1), recipient, network (Sepolia), amount (0.01 SepoliaETH) and estimated network fee (0.0001 SepoliaETH) at market speed of about 12 seconds. After confirming, which signed the transaction with the private key, it was broadcast and appeared as Pending in the Activity tab.

Output:



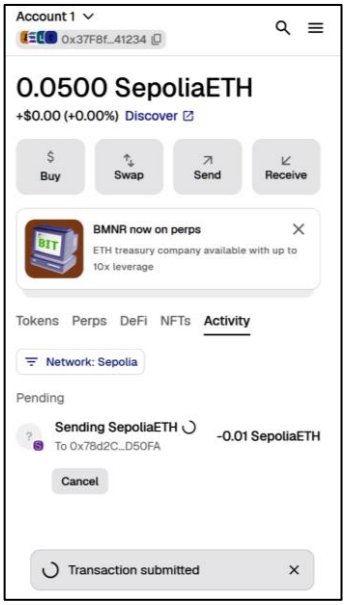
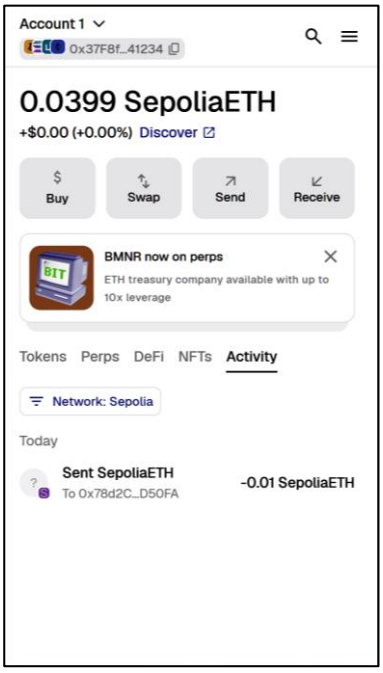


Fig 3 (left): Confirmation screen showing sender, recipient, network, amount and gas fee. Fig 4 (right): Transaction submitted and pending in the Activity tab.

3. Updated Balance and Transaction Receipt

Once the transaction was mined, the wallet balance updated from 0.0500 to 0.0399 SepoliaETH, reflecting the 0.01 SepoliaETH sent plus the gas fee deducted. Opening the entry from the Activity tab displayed the full receipt with a status of Confirmed.

Output:



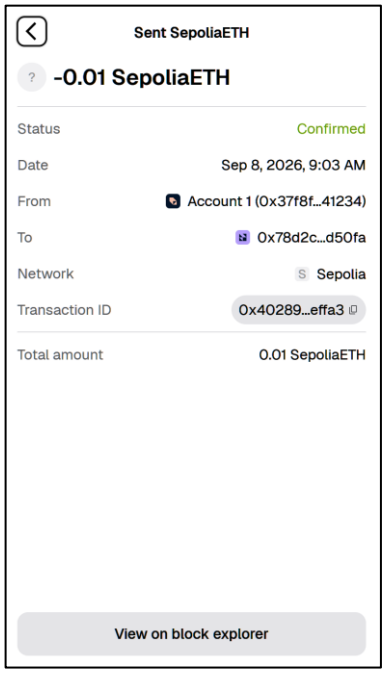


Fig 5 (left): Updated balance of 0.0399 SepoliaETH with the completed transaction. Fig 6 (right): Receipt showing Confirmed status and transaction ID 0x40289...effa3.

4. Verification on the Block Explorer

The transaction was opened on Sepolia Etherscan using its transaction hash. The explorer independently confirmed the transfer of 0.01 ETH, showing a status of Success, inclusion in block 11658562 with 4 block confirmations, and the full sender and recipient addresses.

Output:

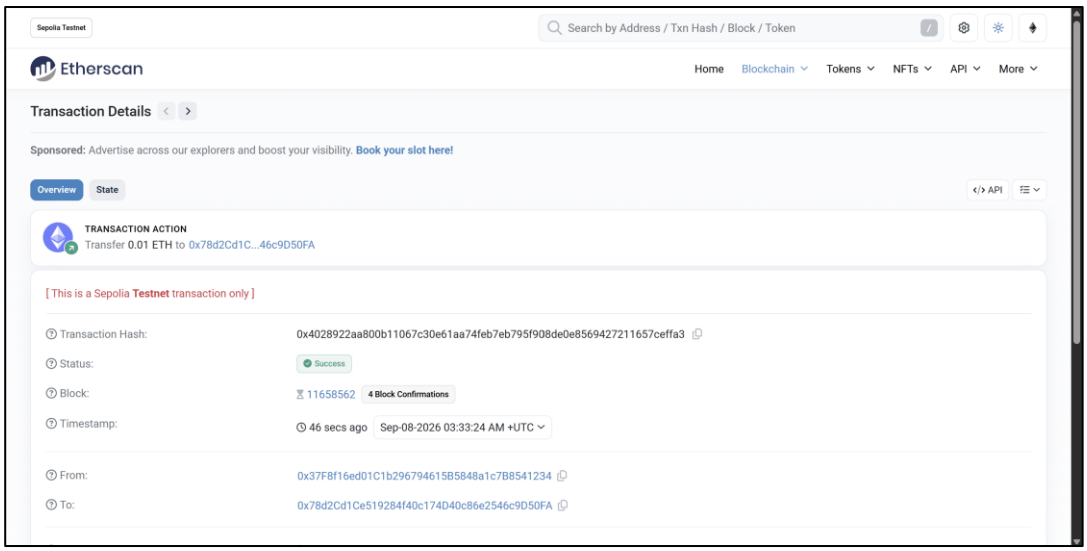


Fig 7: Sepolia Etherscan transaction details confirming the transfer was successfully recorded on the public ledger.

Observations:

1. **The deducted amount exceeded the amount sent.** The balance fell from 0.0500 to 0.0399 SepoliaETH, a reduction of 0.0101 SepoliaETH, while only 0.0100 was transferred to the recipient. The additional 0.0001 SepoliaETH is the network fee paid to validators, showing that every on-chain transaction carries a cost separate from the value being sent.
2. **The transaction passed through three distinct states.** It moved from submitted, to pending in the Activity tab, to confirmed once mined. During the pending stage a Cancel option was available, because the transaction had been broadcast but not yet included in a block.
3. **Confirmation is probabilistic rather than instant.** Etherscan reported 4 block confirmations shortly after the transaction was mined. Each subsequent block added on top makes reversal progressively harder, which is why exchanges typically wait for several confirmations before treating a transfer as final.
4. **The transaction is publicly verifiable without any account.** The same transfer could be independently checked on Etherscan using only the transaction hash. Both the MetaMask receipt and the block explorer report the identical hash, sender, recipient and amount, demonstrating the transparency of a public ledger.
5. **No intermediary was involved at any stage.** The transfer moved directly from one address to another and was validated by the network itself rather than by a bank or payment processor. This is the core peer-to-peer property that distinguishes blockchain transfers from conventional electronic payments.
6. **The testnet removes financial risk while preserving the workflow.** Because SepoliaETH has no monetary value and was obtained free from a faucet, the complete transaction lifecycle could be exercised exactly as it would occur on mainnet, but without any real funds at stake.

Conclusion:

In this experiment a MetaMask wallet account was successfully created and funded with 0.0500 test SepoliaETH on the Sepolia Testnet. A peer-to-peer transaction of 0.01 SepoliaETH was initiated, reviewed and confirmed through MetaMask's Send flow, and the resulting change in account balance from 0.0500 to 0.0399 SepoliaETH correctly accounted for both the amount sent and the 0.0001 SepoliaETH network gas fee.

The transaction was independently verified on Sepolia Etherscan, which reported a status of Success, inclusion in block 11658562 and 4 block confirmations, with the transaction hash 0x4028922aa800b11067c30e61aa74feb7eb795f908de0e8569427211657ceffa3 matching the receipt shown in the wallet. The experiment therefore demonstrated the fundamental blockchain workflow of wallet-based peer-to-peer value transfer without any intermediary, illustrated how gas fees are deducted alongside the transferred amount in every on-chain transaction, and showed how a public block explorer allows any transaction to be independently audited.